

React Concepts Guide

This guide covers the most important React concepts commonly asked in interviews, used in projects, and required for frontend development.

1. What is React?

React is a JavaScript library for building user interfaces. It is component-based, reusable, and efficient because it uses the Virtual DOM.

2. Components

Components are reusable building blocks in React. There are two main types: Functional Components and Class Components. Functional components are preferred in modern React.

3. JSX

JSX stands for JavaScript XML. It allows writing HTML-like syntax inside JavaScript.

4. Props

Props are read-only inputs passed from parent components to child components.

5. State

State stores dynamic data inside a component. Updating state causes the component to re-render.

6. useState Hook

The useState hook is used to add state to functional components.

7. useEffect Hook

useEffect is used for side effects such as API calls, timers, event listeners, and data fetching.

8. Event Handling

React handles events using camelCase syntax such as onClick, onChange, and onSubmit.

9. Conditional Rendering

React can render UI conditionally using if statements, ternary operators, or logical operators.

10. Lists and Keys

Lists are rendered using the `map()` method. Keys help React identify elements efficiently.

11. Forms

Controlled components use React state to manage form inputs.

12. Lifting State Up

When multiple components need the same data, state is moved to their closest common parent.

13. Context API

Context API allows sharing data globally without prop drilling.

14. React Router

React Router enables navigation between pages in single-page applications.

15. API Calls

React commonly uses `fetch()` or `Axios` to retrieve data from backend APIs.

16. Virtual DOM

Virtual DOM improves performance by updating only changed elements instead of reloading the whole page.

17. Hooks

Hooks allow functional components to use React features like state and lifecycle methods.

18. `useRef`

`useRef` provides direct access to DOM elements and persists values without re-rendering.

19. `useMemo` and `useCallback`

These hooks optimize performance by memoizing values and functions.

20. React Lifecycle

Lifecycle methods describe component creation, updates, and destruction. Hooks replace many lifecycle methods in functional components.

Important React Interview Topics

- Difference between state and props
- Difference between PUT and POST
- Controlled vs uncontrolled components
- useEffect dependency array
- React routing
- Hooks in React
- Virtual DOM vs Real DOM
- Redux basics
- Component lifecycle
- API integration in React

Practice building small projects such as Todo Apps, Weather Apps, Authentication Systems, and CRUD Applications to strengthen React skills.